INCH-POUND

MIL-P-50791C (AR)
AMENDMENT 6
10 March 1995
SUPERSEDING
AMENDMENT 5
22 March 1993

MILITARY SPECIFICATION

PROJECTILE, 105MM, TPDS-T, M737
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-50791C (AR), dated 8 May 1987, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.2: American Society for Testing and Materials (ASTM):

Delete: "E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials".

PAGE 4

3.5.2: Delete: "...in a counterclockwise direction (as viewed from the front of the metal parts assembly)".

PAGE 9

Table I Category IV, Line 10: Delete: "...after heat treatment...".

PAGE 16

4.4.2.4: Delete: "102 Hardness 0.40% 3.3 4.5.1.2".

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4.4.2.4: Under Notes, delete:

"After final heat treatment, two bodies from each heat treated lot or six tensile test specimens from every 1,000 ft. of heat treated steel alloy bar, shall be selected at random and subjected to this test."

and substitute:

"Two bodies from each stock material lot or six tensile test specimens from every 1,000 ft. of steel alloy bar, shall be selected at random and subjected to this test."

PAGE 19

4.4.2.5:

Category 110: Delete:

"110	Salt spray test	2	3.6	See Note".

Delete Note: "If any subcaliber assembly fails to meet all the requirements of the drawing and TT-C-490, the lot shall be rejected. The actual subcaliber assembly shall be used in this test."

Category 201: Delete:

"201	Paint thickness	0.65%	3.6	Gage".

PAGE 22

4.4.2.6: Delete:

"204	Location of slots	0.65%	3.1	Gage".

PAGE 24

4.4.2.7:

Category 114: Delete in its entirety.

Category 115: Delete "...max..." and substitute "...min...".

Category 210: Delete in its entirety.

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PAGE 25

4.4.2.7: Category 203: Delete in its entirety.

PAGE 29

4.4.2.10: Delete:

"107 Presence of gap between sabot and base 0.40% 3.1 Gage".

PAGE 37

4.4.3.3.2: Delete:

"4.4.3.3.2 Production lot sample. The quantities of projectile assemblies listed in Table III shall be ballistically tested in the phases indicated.

TABLE III Ballistic test summary.

	TEMP °F	TUBE WEAR	QTY	REQUIREMENT	DEFECT (6.2.4)	ACCEPT	REJECT
A	+70	New	10	a) 3.8.3-Target Impact-Dispersion	Major	4.4.3.3.3	
				b) 3.8.1-Security	Special	0	1
				c) 3.8.2-Flight	Major	0	1
B	+125	Worn	10	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	0 (Note)	2

Note: If one round fails to meet the flight requirement, condition ten additional rounds at 70°F and fire from a worn tube. Reject the lot if in this retest one round doesn't meet the flight requirement."

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and substitute the following:

"4.4.3.3.2 Production lot sample. The quantities of projectile assemblies listed in Table III shall be ballistically tested in the phases indicated.

New producers shall demonstrate manufacturing capability with first article submission and three consecutive lots prior to reduced testing. Current producers that have a break in production shall demonstrate successful completion through first article testing. Phase A is for proven producers whereas, new producers must complete phases A and B.

TABLE III Ballistic test summary.

	TEMP °F	TUBE WEAR	QTY	REQUIREMENT	DEFECT (6.2.4)	ACCEPT	REJECT
A	+70	New	10	a) 3.8.3-Target Impact-Dispersion	Major	4.4.3.3.3	
				b) 3.8.1-Security	Special	0	1
				c) 3.8.2-Flight	Major	0	1
B	+125	Worn	10	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	0 (Note)	2

Note: If one round fails to meet the flight requirement, condition ten additional rounds at 70°F and fire from a worn tube. Reject the lot if in this retest one or more rounds do not meet the flight requirement."

PAGE 38

4.5.1.2: Delete in its entirety.

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PAGE 40

* 4.5.6.1 Delete in its entirety and substitute the following:

"4.5.6.1 Ballistic phases. The metal parts assemblies shall be subjected to ballistic testing at the phases indicated in Table II (First article) and Table III (Production). The metal parts assemblies shall be assembled into complete rounds utilizing the service charge and shall be fired from a fixed-type facility mount with an M1 or M60A3 recoil utilizing a muzzle boresight or from an M1 or M60A3 tank with a gunner's auxiliary sight. An M68A1 gun tube shall be utilized as specified in Table I. All fixed type facility mounts and tank systems must be properly maintained and validated prior to testing."

PAGE 49

Figure 6: Delete "Tape" and substitute "Taped joint - location optional".

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